Nurarys Baibolat CS-2413

Practice Week 9

1. Create a new directory /secure_lab and inside it create 3 files with different owners and permissions. Then list them with detailed info (ls -l).

```
nurarys@nurarys-VirtualBox:~$ sudo mkdir -p /secure_lab
[sudo] password for nurarys:
Sorry, try again.
[sudo] password for nurarys:
nurarys@nurarys-VirtualBox:~$ sudo useradd -m userA
nurarys@nurarys-VirtualBox:~$ sudo useradd -m userB
nurarys@nurarys-VirtualBox:~$ sudo touch /secure_lab/file1 /secure_lab/file2 /secure_lab/file3

nurarys@nurarys-VirtualBox:~$ sudo chown userA /secure_lab/file1
nurarys@nurarys-VirtualBox:~$ sudo chown userB /secure_lab/file2
nurarys@nurarys-VirtualBox:~$ sudo chmod 600 /secure_lab/file1
nurarys@nurarys-VirtualBox:~$ sudo chmod 640 /secure_lab/file2
nurarys@nurarys-VirtualBox:~$ sudo chmod 755 /secure_lab/file3
nurarys@nurarys-VirtualBox:~$ ls -l /secure_lab
total 0
-rw------- 1 userA root 0 Nov  4 08:13 file1
-rw-r----- 1 userB root 0 Nov  4 08:13 file2
-rwxr-xr-x 1 root  root 0 Nov  4 08:13 file3
```

2. Create a shared directory /shared_data where all users can read and write, but only the owner can delete files. Verify with ls -ld.

```
nurarys@nurarys-VirtualBox:~$ sudo mkdir -p /shared_data
nurarys@nurarys-VirtualBox:~$ sudo chmod 1777 /shared_data
nurarys@nurarys-VirtualBox:~$ ls -ld /shared_data
drwxrwxrwt 2 root root 4096 Nov  4 08:15 /shared_data
```

3.Create a new user named admin1 and give them sudo privileges. Then verify that they can run commands with sudo.

```
nurarys@nurarys-VirtualBox:~$ sudo useradd -m -s /bin/bash admin1
nurarys@nurarys-VirtualBox:~$ sudo passwd admin1
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
nurarys@nurarys-VirtualBox:~$ sudo usermod -aG sudo admin1
nurarys@nurarys-VirtualBox:~$ sudo -l -U admin1
Matching Defaults entries for admin1 on nurarys-VirtualBox:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin,
    use_pty

User admin1 may run the following commands on nurarys-VirtualBox:
    (ALL : ALL) ALL
```

4.Set the SUID permission on /usr/bin/passwd and explain what it means. Then find all files in /bin or /usr/bin that have SUID set.

```
nurarys@nurarys-VirtualBox:~$ ls -l /usr/bin/bash
-rwxr-xr-x 1 root root 1446024 Mar 31  2024 /usr/bin/bash
nurarys@nurarys-VirtualBox:~$ sudo chmod u+s /usr/bin/passwd
nurarys@nurarys-VirtualBox:~$ ls -l /usr/bin/bash
-rwxr-xr-x 1 root root 1446024 Mar 31  2024 /usr/bin/bash
nurarys@nurarys-VirtualBox:~$ find /bin /usr/bin -perm -4000 0type f 2>/dev/null
```

5.Create a symbolic link and a hard link to an existing file in /etc (for example, /etc/hostname). Compare inode numbers and explain the difference.

```
nurarys@nurarys-VirtualBox:~$ sudo ln -s /etc/hostname /tmp/hostname_symlink
ln: failed to create symbolic link '/tmp/hostname_symlink': File exists
nurarys@nurarys-VirtualBox:~$ sudo ln /etc/hostname /tmp/hostname_hardlink
ln: failed to create hard link '/tmp/hostname_hardlink': File exists
nurarys@nurarys-VirtualBox:~$ ls -li /ets/hostname /tmp/hostname_symlink /tmp/hostname_hardlink
ls: cannot access '/ets/hostname': No such file or directory
1356573 -rw-r--r-- 2 root root 19 Sep  9 15:33 /tmp/hostname_hardlink
 655523 lrwxrwxrwx 1 root root 13 Nov  4 08:23 /tmp/hostname_symlink -> /etc/hostname
```

6.Change ownership of a file in /secure_lab to another user, and modify permissions so only that user can read and write it. Verify with ls -l.

```
nurarys@nurarys-VirtualBox:~$ sudo chown userB /secure_lab/file3
nurarys@nurarys-VirtualBox:~$ sudo chmod 600 /secure_lab/file3
nurarys@nurarys-VirtualBox:~$ ls -l /secure_lab/file3
-rw------- 1 userB root 0 Nov  4 08:13 /secure_lab/file3
nurarys@nurarys-VirtualBox:~$
```

7.Configure file access control lists (ACLs): give read permission to user student1 on /secure_lab/confidential.txt without changing file ownership. Verify with getfacl.

```
nurarys@nurarys-VirtualBox:~$ sudo touch /secure_lab/confidential.txt
nurarys@nurarys-VirtualBox:~$ sudo setfacl -m u:student1:r /secure_lab/confidential.txt
nurarys@nurarys-VirtualBox:~$ getfacl /secure_lab/confidential.txt
getfacl: Removing leading '/' from absolute path names
# file: secure_lab/confidential.txt
# owner: root
# group: root
user::rw-
user:student1:r--
group::r--
mask::r--
other::r--
```

8.Create a tar archive of /etc with restricted access: only root can read it. Store it as /root/etc_backup.tar.gz and confirm permissions.

```
nurarys@nurarys-VirtualBox:~$ sudo tar -czf /root/etc_backup.tar.gz /etc
tar: Removing leading `/' from member names
tar: Removing leading `/' from hard link targets
nurarys@nurarys-VirtualBox:~$ sudo chmod 600 /root/etc_backup.tar.gz
nurarys@nurarys-VirtualBox:~$ ls -l /root/etc_backup.tar.gz
ls: cannot access '/root/etc_backup.tar.gz': Permission denied
nurarys@nurarys-VirtualBox:~$ sudo ls -l /root/etc_backup.tar.gz
-rw------- 1 root root 1275457 Nov  4 08:34 /root/etc_backup.tar.gz
```

9.Use chmod, chown, and umask to create a new default permission policy where new files get rw-r----- permissions by default. Show verification with umask.

```
nurarys@nurarys-VirtualBox:~$ umask 0027
nurarys@nurarys-VirtualBox:~$ umask
0027
nurarys@nurarys-VirtualBox:~$ touch testfile && ls -l testfile && rm testfile
-rw-r----- 1 nurarys nurarys 0 Nov  4 08:39 testfile
```

10. List all users who have a valid shell (not /usr/sbin/nologin) and redirect the result to a file named login_users.txt.

```
nurarys@nurarys-VirtualBox:~$ grep -E -v '(/usr/sbin/notlogin|/bin/false)' /etc/passwd | cut -d:
 -f1 > ~/login_user.txt
nurarys@nurarys-VirtualBox:~$ cat ~/login_user.txt
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
```